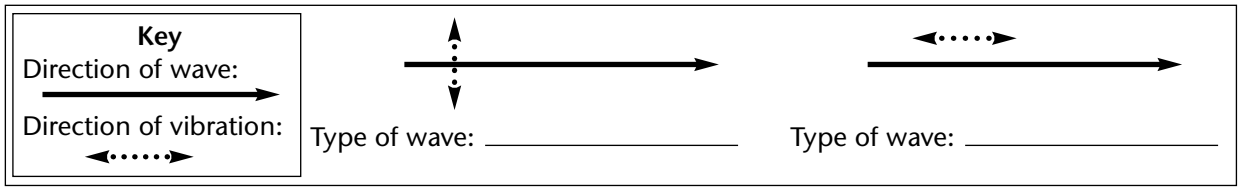
Types of waves

4a

Waves involve vibrations. They carry energy from one place to another. There are two types of waves: longitudinal (lengthwise) waves in which the medium carrying the wave vibrates along the direction in which the wave is travelling, and transverse (across) waves in which the vibrations are perpendicular to the direction of wave movement. Longitudinal waves are sometimes called compression waves.

1 Use the spaces provided to label each wave diagram.



2 One type of wave (sound waves are of this type) always consists of vibrating particles that pass energy on to other particles, which then start to vibrate themselves. State which of the above wave types could be a sound wave. Justify your answer.

_____

3 If the energy of vibrating particles is passed on from particle to particle, in which medium would sound travel faster: air, water or iron? Choose a medium and justify your answer.

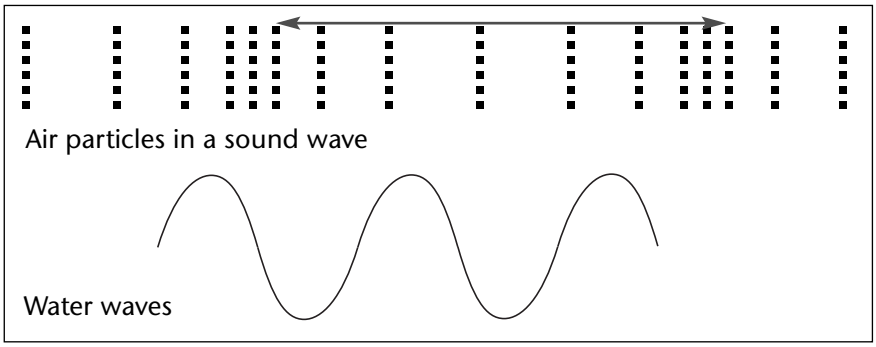
_____

4 Suggest what happens to the speed of sound at higher levels in the earth's atmosphere. Justify your answer.

_____

5 Explain why sound does not travel in space. _____

6 In addition to vibration patterns, waves can differ in wavelength and frequency. Generated by vibrating sources, waves will have the same frequency as their source and the pattern of disturbance will repeat itself regularly. A wavelength is the distance between any two closest points along the wave for which the disturbances are in step. Indicate three wavelengths on each diagram at right. One wavelength has been drawn as an example.



7 **a** Amplitude refers to the range of movement of the vibration in a wave. It is about loudness or intensity. In longitudinal waves the amplitude is measured from the centre of a compression to the centre of the closest rarefaction. Circle and label the middle of a compression and the middle of the rarefaction of the sound wave above.

b Measure the amplitude of that sound wave in mm. _____

8 **a** In transverse waves the amplitude goes from the zero line (i.e. the line around which the disturbance vibrates) to the top of a crest (or bottom of a trough). Circle and label a crest and a trough in the transverse wave above. Then draw the zero line.

b Measure the amplitude of the water wave in mm. _____

The frequency of a wave measures the number of cycles of the disturbance or the number of vibrations that take place in one second. For a travelling wave, the number of wavelengths that pass a point in one second is also a measure of the frequency. The unit of frequency is the hertz; e.g. 2 Hz (i.e. two hertz) is two cycles per second. Frequency is calculated by dividing the number of cycles or waves by the time they took to travel past one point (i.e. 2 waves/1 s = 2 Hz). (In general, if the wavelength is big, then the

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Types of waves

4b

frequency is _____.)

9 Calculate the frequencies of the musical notes listed in the table below. (When talking about music and the human ear the word pitch is often used rather than frequency.) Note A above the middle C is known as concert pitch.

Note	Number of cycles	Time (s)	Calculation	Frequency (Hz)
Middle C	256	1		
A	880	2		
D	864	3		
High C	2048	4		
E	1400	5		
F	341	1		
G	678	2		
A	660	3		
B	960	4		
A	550	5		
B	960	2		

10 List the notes in order of frequency (low to high) and select eight consecutive notes to form an octave. Start with A (220 Hz) and list the immediately following notes in order.

_____

11 List the frequencies of all A notes.

_____

12 Write a generalisation about the relationship between any two adjoining A notes.

_____

13 The speed of a wave depends on the medium it is travelling through. Light travels through empty space (i.e. a vacuum) at 300 000 000 m/s (more accurately 299 792 458 m/s), through glass at 200 000 000 m/s, through diamond at 125 000 000 m/s and through water at 225 000 000 m/s. Light speed through air is practically the same as light speed in a vacuum. Record these speeds in the table (highest to lowest).

Medium				
Speed of light				

14 Sound travels through air at 20°C at a speed of 342 m/s. This speed is about one million times slower than the speed of light. When air temperature drops to 0°C, sound speed drops to 331 m/s. Water carries sound at a speed of 1 500 m/s, brick at 3 600 m/s, wood at 4 000 m/s and iron at 6 000 m/s. Suggest a reason why sound speed drops, when the air temperature drops. (Think of the kinetic particle theory.)

_____

15 Make a generalisation about how the speed of sound changes, as the density of the transmitting medium changes.

_____

16 Relate the speed of light to the density of the transmitting medium.

_____

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